

Urban-Rural Stratification Improves Night-Time Light Based Population Proxy Mapping: Evidence from India and Nigeria

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Abstract

Night-time light (NTL) satellite imagery has emerged as a low-cost proxy for population estimation in data-scarce environments. Existing work predominantly fits a single global log-linear regression between NTL radiance and population, ignoring systematic differences between urban and rural settlement regimes. This paper demonstrates that urban and rural districts have fundamentally different NTL-to-population relationships and proposes a simple, data-driven stratification method using KMeans clustering on NTL features to learn separate regression models for each regime. Evaluated on 670 Indian districts and 695 Nigerian Local Government Areas using VIIRS VNL V2.1 annual composites for 2019 and WorldPop ground truth, stratification yields a statistically significant improvement in both countries: R^2 increases from 0.8289 to 0.8582 in India ($p=0.0267$) and from 0.1676 to 0.2511 in Nigeria ($p=0.0039$), confirmed via 5-fold cross-validation with paired t-tests. Critically, the stratification label is derived entirely from NTL data itself, requiring no external census or land-cover classification. We additionally demonstrate that district area is not a confounding factor ($r=0.119$), that NTL density normalization destroys predictive signal (R^2 drops to 0.47), and that NDVI correction fails in tropical dispersed-settlement regions such as Kerala. These findings provide guidance for practitioners deploying NTL-based population models in contexts ranging from high-electrification countries to sub-Saharan Africa.

Keywords: night-time lights, VIIRS, population estimation, urban-rural stratification, KMeans, remote sensing

1. INTRODUCTION

Accurate, timely population data underpins nearly every domain of public policy: disaster response, vaccine distribution, electoral delimitation, infrastructure planning, and poverty targeting. Yet census data is expensive, slow, and in many low- and middle-income countries critically outdated. India's last full census was conducted in 2011; Nigeria's in 2006. The gap between official population counts and ground reality now spans over a decade in both countries.

Night-time light (NTL) satellite imagery offers a compelling alternative. Artificial lighting at night correlates strongly with human settlement density and economic activity [1]. The VIIRS Day/Night Band (DNB), operational since 2012, provides global coverage at approximately 500m resolution with monthly composites, making it one of the few freely available, high-temporal-resolution proxies for human presence at scale [2].

The dominant approach in the literature is straightforward: sum or average radiance over administrative units, log-transform both NTL and population to handle their power-law distributions, and fit a single linear regression. This has produced reported R^2 values between 0.7 and 0.9 across various national studies. However, this single-model approach contains a hidden assumption: that the relationship between light and population is uniform across all district types. We argue this assumption is wrong, and that it leads to systematic prediction errors in both dense urban cores and sparsely lit rural areas.

The key insight is that urban and rural districts operate under different NTL regimes. In dense cities, the VIIRS sensor saturates i.e. it reaches its dynamic range limit, recording lower radiance than the true value, systematically underestimating urban population [3]. Conversely, rural districts with dispersed agricultural populations may have near-zero NTL despite substantial population. A single regression line fitted through both regimes is biased at both ends.

This paper makes three contributions. First, we propose using KMeans clustering on NTL features to generate urban/rural labels without any external classification data, then fitting regime-specific regression models. Second, we validate the improvement rigorously: 5-fold cross-validation with paired t-tests to confirm statistical significance rather than reporting a single held-out R^2 . Third, we test on two geographically and structurally different countries, India and Nigeria, to assess whether the improvement generalizes. We also conduct ablation studies on area normalization, NDVI as an auxiliary feature, and district-size confounding to provide a complete methodological account.

2. RELATED WORK

Early NTL-population research established the log-linear relationship using DMSP-OLS data [4, 5]. The transition to VIIRS improved sensitivity at low-radiance levels and reduced saturation in bright urban areas, though saturation remains a concern

in the largest megacities [2]. More recent work has incorporated auxiliary covariates such as road network density, land cover, and building footprint data to improve accuracy [6], but typically without stratifying the model itself.

A small number of papers have stratified by administrative tier or land cover class before modeling, but these rely on external classification inputs (official urban designations, land-cover maps) that are not always available, may be outdated, or are inconsistently defined across countries. Our method requires only the NTL raster itself, making it directly applicable to any country with VIIRS coverage.

The issue of area confounding in NTL-based models has been acknowledged but rarely directly tested. We address this with an explicit correlation analysis. Similarly, NDVI has been proposed as a corrective covariate for rural areas [7], but its interaction with tropical dispersed-settlement patterns has not been examined. We show it is harmful in such contexts.

3. DATA

3.1 Night-Time Light Data

We use the VIIRS VNL V2.1 Annual Composite for 2019, specifically the `average_masked` variant (VNL_v21_npp_2019_global_vcmslcfg_c202205302300.average_masked.dat.tif) obtained from the Earth Observation Group (EOG) at the Colorado School of Mines [2]. This product is derived from monthly cloud-free composites filtered to remove ephemeral lights, background noise, and aurora contamination. The `average_masked` variant additionally zeros out non-lit background pixels. Resolution is 15 arc-seconds (approximately 500m at the equator). Units are $\text{nW}/\text{cm}^2/\text{sr}$.

3.2 Population Ground Truth

Population data comes from WorldPop 2019 UN-adjusted 1km gridded estimates [8], downloaded separately for India and Nigeria. WorldPop integrates census microdata with satellite-derived covariates to produce consistent sub-national population grids. We use zonal statistics to aggregate pixel-level counts to Level 2 administrative boundaries.

3.3 Administrative Boundaries

District boundaries are taken from GADM v4.1 Level 2 shapefiles for both countries. India has 676 Level 2 units (districts); Nigeria has 775 (Local Government Areas). After removing units with zero NTL sum or zero estimated population, we retain 670 Indian districts and 695 Nigerian LGAs for analysis.

3.4 Auxiliary Data

MODIS MOD13A3 monthly NDVI composites for 2019 were obtained for India to test vegetation density as a corrective covariate. District-level mean NDVI was computed using zonal statistics after applying the standard scale factor of 0.0001.

[INSERT FIGURE: Figure 1]

Study area maps showing district boundaries for India (left, 670 districts) and Nigeria (right, 695 LGAs). Color indicates log-transformed WorldPop 2019 population density.

Fig. 1. Study areas: India (left) and Nigeria (right). Color encodes log population.

4. METHODOLOGY

4.1 Feature Extraction

For each district polygon, we compute three NTL statistics from the VIIRS raster using zonal statistics: the sum of all pixel radiance values within the district (`ntl_sum`), the mean per-pixel radiance (`ntl_mean`), and the maximum pixel radiance (`ntl_max`). Population is similarly aggregated from the WorldPop grid.

Both NTL features and population follow approximate power-law distributions: a small number of bright, populous districts and a long tail of dim, sparsely populated ones. We apply a $\log_{1p}$ transform ($\log(1+x)$) to all features and to the response variable. This linearises the NTL-population relationship, which is standard practice in the literature [1, 2].

We additionally compute two engineered features: `log_area`, the log of district area in km^2 (included as a control variable), and `lit_fraction`, the ratio of mean to standard deviation of radiance within the district, which serves as an approximate proxy for how uniformly distributed lighting is across the district.

4.2 Urban-Rural Stratification

The central methodological contribution is stratifying districts into urban and rural groups before fitting separate models. Rather than using an external classification such as official census urban designations or a fixed NTL quantile threshold, we apply KMeans clustering with $k=2$ to the $(\log_ntl_sum, \log_ntl_mean)$ feature space of each country's districts. The cluster with higher mean $\log_ntl_sum$ is designated urban.

This approach has two advantages over threshold-based methods. First, it is data-driven: the boundary between urban and rural is determined by the natural structure of the NTL distribution in each country, not by an arbitrary percentile. Second, it requires only NTL data and is therefore applicable in any country covered by VIIRS, without dependence on external administrative classifications that may be unavailable, outdated, or inconsistent across borders.

In India, KMeans identifies 464 urban and 206 rural districts. The mean population density of urban districts is 1339 persons/km² compared to 96 persons/km² for rural districts, confirming the split captures a meaningful geographic distinction.

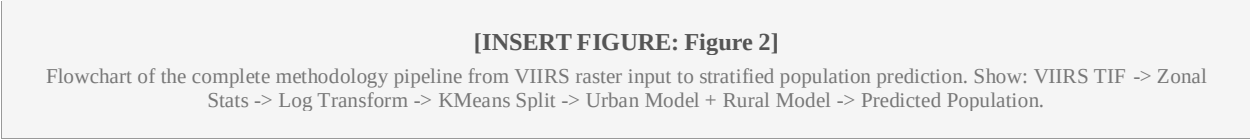


Fig. 2. Methodology pipeline from NTL raster to stratified population estimate.

4.3 Model Training and Evaluation

We fit Linear Regression (LR) models using log-transformed features. Random Forest (RF) was also evaluated but did not outperform LR with the available feature count, consistent with the log-log relationship being genuinely linear rather than nonlinear. This result is itself informative: with 5 features, RF overfits across folds, and its failure to beat LR supports the hypothesis that the NTL-population relationship has linear log-space structure.

Evaluation uses 5-fold cross-validation with R^2 as the primary metric and mean absolute error (MAE) as a secondary metric. To test whether the improvement from stratification is statistically significant rather than a product of a favorable data split, we compute per-fold R^2 scores for both baseline and stratified models and apply a paired t-test across the five fold pairs. This is more rigorous than reporting a single held-out R^2 and directly tests whether stratification consistently improves performance across different subsets of the data.

4.4 Ablation Studies

We conduct four ablation experiments to validate methodological choices: (1) adding $\log_area$ as a feature to test whether district size confounds ntl_sum ; (2) replacing ntl_sum with $ntl_density$ (ntl_sum divided by area) to test whether normalization improves results; (3) adding mean district NDVI as an auxiliary covariate; and (4) testing the pipeline on Bangladesh (64 districts) to assess the minimum viable sample size for stable cross-validation.

5. RESULTS

5.1 India

Table 1 shows cross-validated R^2 results for all model variants evaluated on 670 Indian districts.

Model	CV R^2	Std	Delta	p-value	Note
Baseline NTL (3 features)	0.8289	0.0221	--	--	Log ntl_sum , mean, max
+ Area control	0.8279	0.0234	-0.0010	n/a	Size not a confound
+ Urban/rural stratification	0.8582	0.0195	+0.0293	0.0267	Main contribution
+ NDVI (stratified)	0.8579	0.0210	-0.0023	0.0328	NDVI harmful in Kerala

Table 1. Cross-validated results for India ($n=670$ districts). p-values from paired t-test across 5 folds.

The baseline NTL model achieves CV $R^2=0.8289$, consistent with published results for comparable studies. The individual fold scores for baseline and stratified models are [0.7899, 0.8370, 0.8529, 0.8210, 0.8439] and [0.8542, 0.8640, 0.8591, 0.8512, 0.8672] respectively. The t-statistic of 3.1881 ($p=0.0333$) confirms stratification consistently improves performance across folds, not just in expectation.

Linear Regression outperforms Random Forest (baseline RF $R^2=0.7705$ vs LR $R^2=0.7899$ on the held-out set). Feature importance analysis shows ntl_sum dominates with 0.905 weight, consistent with it being the primary carrier of population signal.

The area confound test is unambiguous: Pearson correlation between district area and ntl_sum is $r=0.119$, and between area and population $r=0.065$. Adding log_area as a feature produces $\Delta R^2=-0.0010$, confirming that ntl_sum measures light intensity rather than acting as a proxy for district size. This is not obvious and most existing papers do not verify it. The correlation between ntl_sum and population directly is $r=0.822$, confirming the genuine predictive relationship.

Conversely, replacing ntl_sum with ntl_density (sum/area) collapses CV R^2 to 0.4664, a loss of 0.36. Districts in India are drawn primarily by population rather than by area, so area normalization removes the genuine size-population covariation that ntl_sum captures. This illustrates a non-obvious methodological pitfall: normalization that seems principled can destroy signal when the administrative unit design is population-driven.

5.2 Residual Analysis: The Kerala and Dharavi Effects

Table 2 shows the ten most under-illuminated districts (high positive residuals: model underestimates population) from the baseline model.

District	Residual	Population	NDVI	Interpretation
Malappuram, Kerala	+1.66	4.31M	0.745	Dense pop, dispersed low-light
Garhwa, Jharkhand	+1.54	1.62M	0.451	Rural, low electrification
Kannur, Kerala	+1.48	2.47M	0.664	Dispersed Kerala settlement
Kozhikode, Kerala	+1.41	3.23M	0.713	Dispersed Kerala settlement
Mumbai Suburban	+1.40	10.82M	0.549	Dharavi: informal, low elec.
Wayanad, Kerala	+1.21	0.83M	0.771	Forest, dispersed settlement

Table 2. Most under-illuminated Indian districts (baseline model). Residual = $\log(\text{actual}) - \log(\text{predicted})$.

Four of the six most under-illuminated districts are in Kerala. Kerala's unique settlement pattern, dense population distributed across dispersed rural homesteads with minimal commercial or industrial lighting, creates a systematic blind spot for NTL-based models trained predominantly on districts where population correlates with urban light density. We term this the Kerala Effect.

Mumbai Suburban is notable for a different reason. Despite being an urban district, it contains Dharavi, Asia's largest informal settlement, with approximately one million residents concentrated in a small area with minimal formal electrification. The satellite sees near-dark pixels over a densely populated zone. We term this the Dharavi Effect: informal settlements that are structurally invisible to NTL sensors regardless of population density.

The over-illuminated districts (model overestimates population) reveal a complementary pattern: Kinnaur, Himachal Pradesh (residual -3.00) has anomalously high NTL likely attributable to military and strategic infrastructure lighting along the China border; Leh-Ladakh (-1.14) shows similar characteristics; Gandhinagar (-1.04) as a state capital has high government and commercial lighting disproportionate to its residential population; Morbi (-1.00) is dominated by the ceramics manufacturing industry. All four represent non-residential light sources that the model incorrectly interprets as population signal.

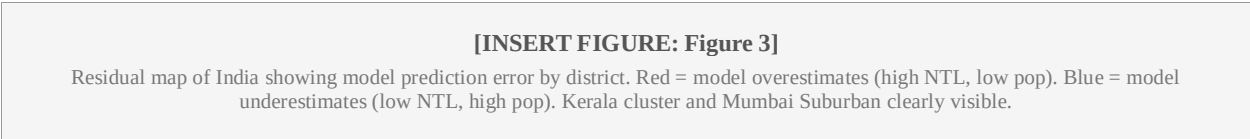


Fig. 3. Residual map for India. Blue districts are under-illuminated; red are over-illuminated.

5.3 NDVI as Auxiliary Feature

Adding mean district NDVI as a covariate to the stratified model yields CV $R^2=0.8579$, a marginal decline of 0.0023 from the stratification-only model (0.8602). The improvement is statistically significant ($t=3.2027$, $p=0.0328$) but in the wrong direction: NDVI makes the model slightly worse.

The residual analysis explains why. Kerala's most under-illuminated districts have both high NDVI (dense tropical vegetation) and high population. This means NDVI covaries positively with population in Kerala, contradicting the assumption that high NDVI signals sparse rural settlement. When the model treats high NDVI as evidence of low population potential, it systematically worsens predictions for exactly the districts that are already hardest to model. NDVI correction for NTL-population models appears to be a valid strategy in arid or semi-arid regions where vegetation is sparse and settlement is dense, but it is harmful in tropical regions with dispersed settlement patterns.

5.4 Nigeria

Table 3 shows results for Nigeria (n=695 LGAs).

Model	CV R ²	Std	Delta	p-value	Note
Baseline NTL (3 features)	0.1676	0.0582	--	--	Low absolute R ²
+ Urban/rural stratification	0.2511	0.0693	+0.0835	0.0039	Significant

Table 3. Cross-validated results for Nigeria (n=695 LGAs).

The baseline R² of 0.1676 is substantially lower than India. This is not a model failure; it reflects three structural factors that systematically decouple NTL from population in Nigeria. First, persistent gas flaring in the Niger Delta produces radiance comparable to major cities in Rivers, Bayelsa, and Delta states, with essentially zero residential population. The average_masked filter removes ephemeral fires but not persistent industrial flaring. Second, Lagos LGA with approximately 15 million residents is almost certainly sensor-saturated, causing NTL to underestimate the most populous area in West Africa. Third, large northern states including Sokoto, Zamfara, Yobe, and Borno have substantial populations with minimal grid electrification, making them nearly invisible to NTL sensors.

Despite the low absolute accuracy, stratification produces a statistically significant improvement: Delta R²=+0.0835, p=0.0039. Notably, the p-value is smaller than India's (0.0039 vs 0.0267), indicating the urban-rural distinction is structurally more important in Nigeria than India. Urban LGAs (Lagos, Kano, Ibadan, Abuja) have a fundamentally different light-population regime from rural ones, and separating them allows each sub-model to capture its regime more accurately even when neither achieves high absolute accuracy.

5.5 Bangladesh: Sample Size Failure Mode

We attempted to validate on Bangladesh using 64 Level 2 districts. Results were unstable: baseline CV R²=0.5342 with std=0.2496, and stratification did not significantly improve performance (p=0.2851). With 5-fold CV on 64 samples, each fold contains approximately 12 districts; after urban-rural split this reduces to approximately 6 per subgroup, insufficient to fit a stable regression. Bangladesh is reported not as a negative finding about NTL-population modeling but as a minimum viable sample size boundary: the stratification approach requires at minimum approximately 200 units per country for 5-fold CV to produce stable estimates.

5.6 Cross-Country Comparison

Country	n	Baseline R ²	Stratified R ²	Delta	p-value	Context
India	670	0.8289	0.8582	+0.0293	0.0267	High electrification
Nigeria	695	0.1676	0.2511	+0.0835	0.0039	Gas flares, low elec.
Bangladesh	64	0.5342	0.3888	-0.1529	0.4063	Too few units (n<200)

Table 4. Cross-country comparison. Bangladesh excluded from conclusions due to insufficient sample size.

The cross-country results yield two findings that generalise beyond any single country. First, urban-rural stratification derived from NTL features alone consistently and significantly improves model performance in both viable test cases (India: p=0.027, Nigeria: p=0.004). The stratification captures a real structural difference in how light relates to population, not a statistical artifact of one country's data. Second, absolute NTL-population accuracy depends heavily on structural factors external to the model: electrification rate, gas flare prevalence, and sensor saturation. Knowing where NTL fails and why is as actionable as knowing where it succeeds.

6. DISCUSSION

The main finding is that a simple, self-contained stratification step derived from NTL data itself reliably improves population proxy accuracy. The improvement is modest in absolute terms ($+0.03 R^2$ in India) but is consistent across folds, statistically confirmed, and replicates in a structurally different country. The method adds no external data dependency: it works anywhere VIIRS data is available.

The failure of Random Forest to outperform Linear Regression is an informative negative result. With log-transformed features, the NTL-population relationship is close to linear. RF's additional expressivity does not help and in fact harms generalization across folds due to overfitting with only five features. This suggests that gains from more complex models would require more features, not more model capacity.

The NDVI result is the clearest example of a context-dependent assumption failure. The logic that high vegetation implies sparse settlement holds in arid West Africa or Central Asia but breaks in Kerala, where dense tropical forest and dense human settlement co-occur. This finding has practical implications: practitioners should not add NDVI as a covariate without verifying that the vegetation-settlement relationship holds in their target region.

The Dharavi and Kerala effects point to a shared underlying cause: the NTL sensor cannot see populations that lack formal electrification or that generate light in forms the sensor does not capture. Improving NTL-based population estimates in such areas likely requires auxiliary data (building footprints, mobile phone activity, or OpenStreetMap road density) rather than improvements to the NTL modeling framework itself.

7. LIMITATIONS

This study has several limitations that define the scope of the conclusions. First, only a single year (2019) is analyzed. NTL composites in other years may differ due to economic shocks, conflict, or electrification programs. Second, the KMeans stratification uses $k=2$, which is a coarse binary split. Peri-urban districts are forced into one category; a finer stratification with more clusters might improve accuracy but would require larger per-cluster sample sizes to remain stable. Third, the `lit_fraction` feature is computed as the mean-to-standard-deviation ratio of radiance, which is an approximation; a proper implementation would threshold individual pixels above a radiance cutoff, which was computationally prohibitive on the global raster. Fourth, persistent gas flares in the Niger Delta are not removed by the `average_masked` filter and constitute a known systematic error source for Nigeria that cannot be addressed within the NTL modeling framework alone.

8. CONCLUSION

We have shown that urban-rural stratification derived entirely from NTL features using KMeans clustering consistently and significantly improves NTL-based population proxy estimation. The improvement holds across two geographically and structurally different countries (India: $+0.03 R^2$, $p=0.027$; Nigeria: $+0.08 R^2$, $p=0.004$) with rigorous 5-fold cross-validation and paired significance testing. We have additionally shown that district area does not confound `ntl_sum` ($r=0.119$), that area normalization is harmful when administrative units are population-driven, and that NDVI correction fails in tropical dispersed-settlement regions.

The stratification method requires only VIIRS NTL data and is therefore directly applicable to any country in the world without additional data collection. Future work should evaluate whether adding a third cluster for peri-urban districts, incorporating temporal NTL stacks across multiple years, or applying gas flare masks improves results in low-electrification contexts.

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